04 — The Hamilton–Jacobi–Bellman Equation and Verification

Goal of this note. Turn the dynamic programming principle of note 03 into a PDE, the **Hamilton–Jacobi–Bellman (HJB) equation**, and prove the **verification theorem**: a smooth solution of HJB with the right growth *is* the value function, and its pointwise maximizer *is* an optimal feedback control. Then extend both to controlled **jump** intensities, which is the form used for Avellaneda–Stoikov (note 12).

The logic is the same as note 02, with one sup added:

Note 02 (no control)	This note
DPP without control (3.1) + Dynkin $\Longrightarrow \partial_t u + \mathcal{L}u = 0$ (§3)	DPP (note 03, 2.1) + Dynkin $\Longrightarrow \partial_t v + \sup_a\{\mathcal{L}^a v + f\} = 0$ (§3.1)
PDE solution $\Longrightarrow$ expectation, via Itô + martingale (§2)	HJB solution $\Longrightarrow$ value function, via Itô + supermartingale (§3.3)

Prerequisites. Note 01: Itô (5.1)–(5.2), jumps (6.3)–(6.4), Dynkin (7.2), the local-martingale caution after (3.2). Note 02: Feynman–Kac (1.3), proof steps (2.1)–(2.4). Note 03: setup (1.1)–(1.7), (H1)–(H4), Lemma 1.2, DPP (2.1)–(2.3), Corollary 2.2, the example of §4.

Notation

Symbol	Meaning	Type
X_s, α_s, A, $\mathcal{A}_t$	state, control, control set, admissible controls (note 03 §1)	—
b, σ, f, g	drift, volatility, running reward, terminal reward (note 03 (1.1), (1.6))	functions
J, v	reward of a control, value function (note 03 (1.6)–(1.7))	functions
$\mathcal{L}^a$	generator of the state when the control is frozen at $a \in A$	operator
p, M	placeholders for $\partial_x v$ (gradient) and $\partial_{xx} v$ (Hessian)	$\mathbb{R}^d$, $d \times d$ symmetric
$H(t,x,p,M)$	Hamiltonian, eq. (1.2)	real
w	candidate solution of the HJB equation	function
$\hat{a}(t,x)$	maximizer in the Hamiltonian: candidate optimal feedback	function $\to A$
τ_n	localizing stopping times in the verification proof	stopping times

Symbol	Meaning	Type
$\mathcal{K}_n$, C_n	compact set visited before τ_n, bound of w and its derivatives on it	set, constant
$G(s,y)$	$\partial_s v + H(s,y,\partial_x v,\partial_{xx} v)$ in §3.1	function
$\bar\lambda$, $\bar\eta$, $\bar\delta$	bounds on intensities, jump sizes, quote distances	constants
$\theta(t,q)$	non-trivial part of the market maker's value, eq. (4.7)	function
N^i_s, λ_i, η_i	counting processes, their controlled intensities, jump sizes	—
x, q, s (in §3.5, §4.2)	market maker's cash, inventory (integer), mid price	$\mathbb{R}$, $\mathbb{Z}$, $\mathbb{R}$
δ^b, δ^a	distances of bid and ask quotes from mid (the controls)	≥ 0
$\Lambda(\delta) = Ae^{-k\delta}$	fill intensity of a quote at distance δ	> 0
ϕ	running inventory penalty	≥ 0
c, σ, $P(t)$, $K(t)$	LQ example parameters and coefficients (note 03 §4)	—
$\tau = T - t$	time to maturity	≥ 0

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1. Setup & definitions

We keep the setting of note 03: state equation (1.1) there, admissible controls Def. 1.1, reward (1.6), value (1.7), assumptions (H1)–(H4). Start in $d = 1$, $m = 1$ for readability; the d-dimensional formulas follow by note 01 eq. (5.2).

Definition 1.1 (Controlled generator). For a fixed control value $a \in A$ and $\varphi \in C^{1,2}$,

$$\mathcal{L}^a\varphi(t,x) = b(t,x,a)\,\partial_x\varphi(t,x) + \tfrac{1}{2}\sigma^2(t,x,a)\,\partial_{xx}\varphi(t,x). \tag{1.1}$$

In d dimensions: $\mathcal{L}^a\varphi = b\cdot\nabla\varphi + \frac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top\nabla^2\varphi\big)$. This is note 01's generator (7.1) with the coefficients evaluated at control value a. By Itô's lemma (note 01 eq. 5.1) applied to (note 03 eq. 1.1), for **any** admissible control,

$$d\varphi(s,X_s) = \big(\partial_s\varphi + \mathcal{L}^{\alpha_s}\varphi\big)(s,X_s)\,ds + \sigma(s,X_s,\alpha_s)\,\partial_x\varphi(s,X_s)\,dW_s.$$

Itô's lemma applies because X is an Itô process with adapted coefficients $b_s = b\big(s,X_s,\alpha_s\big)$, $\sigma_s = \sigma\big(s,X_s,\alpha_s\big)$; the control does not need to be a feedback.

Definition 1.2 (Hamiltonian). For $(t,x,p,M) \in [0,T]\times\mathbb{R}\times\mathbb{R}\times\mathbb{R}$,

$$H(t,x,p,M) = \sup_{a\in A}\Big\{b(t,x,a)\,p + \tfrac{1}{2}\sigma^2(t,x,a)\,M + f(t,x,a)\Big\}. \tag{1.2}$$

In d dimensions, $p\in\mathbb{R}^d$, M symmetric $d\times d$, and $\frac{1}{2}\sigma^2 M$ becomes $\frac{1}{2}\operatorname{tr}(\sigma\sigma^\top M)$.

Two remarks on (1.2):

- H is a **supremum of functions affine in** (p, M), hence **convex** in (p, M). It is nondecreasing in M (since $\sigma^2 \geq 0$): this is *degenerate ellipticity*, the structural property behind the comparison principle of note 05.
- H may be $+\infty$ (e.g. $A = \mathbb{R}$, $b = a$, $f = 0$: the controller can push the state at no cost). Then the problem is ill-posed. Under (H4) (compact A) and continuity, H is finite and continuous.

The name comes from classical mechanics: with f a negative kinetic energy $-\frac{1}{2}a^2$ and $b = a$, $H(p) = \sup_a\{ap - \frac{1}{2}a^2\} = \frac{1}{2}p^2$ is the Legendre transform, as in Hamilton's equations.

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2. Statement

2.1 The HJB equation

HJB equation. The value function is expected to satisfy

$$\boxed{\partial_t v(t,x) + H\big(t, x, \partial_x v(t,x), \partial_{xx} v(t,x)\big) = 0 \ \text{ on } [0,T) \times \mathbb{R}, \qquad v(T,x) = g(x).} \tag{2.1}$$

Written out: $\partial_t v + \sup_{a \in A}\{\mathcal{L}^a v + f(\cdot, a)\} = 0$.

Compare note 02 eq. (1.2) with $r = 0$: $\partial_t u + \mathcal{L} u + f = 0$. The HJB is that linear equation with a **pointwise** sup over the control value. It is a fully nonlinear (in general) backward parabolic equation; it is linear iff A is a single point.

Feedback from the Hamiltonian. If v is smooth and the sup in (2.1) is attained, define

$$\hat{a}(t,x) \in \arg\max_{a \in A} \Big\{\mathcal{L}^a v(t,x) + f(t,x,a)\Big\}. \tag{2.2}$$

The infinite-dimensional optimization over processes (note 03 eq. 1.7) has become a **finite-dimensional optimization at each point** (t,x). This is the main practical payoff of dynamic programming.

2.2 Verification theorem

Theorem 2.1 (Verification). Let $(t,x) \in [0,T) \times \mathbb{R}^d$. Assume (H1)–(H2) of note 03 and:

- **(V1) Regularity and growth.** $w \in C^{1,2}([0,T) \times \mathbb{R}^d) \cap C([0,T] \times \mathbb{R}^d)$ and, for some C_w, $p \geq 1$: $|w(s,y)| \leq C_w(1 + |y|^p)$.
- **(V2) Supersolution.** For all $(s,y,a) \in [0,T) \times \mathbb{R}^d \times A$,

$$\partial_s w(s,y) + \mathcal{L}^a w(s,y) + f(s,y,a) \leq 0, \qquad \text{and} \qquad w(T,y) \geq g(y). \tag{2.3}$$

- **(V3) Integrability of admissible controls.** Every $\alpha \in \mathcal{A}_t$ satisfies, with the same p,

$$\mathbb{E}\Big[\sup_{t \leq s \leq T} |X_s^{t,x,\alpha}|^p\Big] < \infty \qquad \text{and} \qquad \mathbb{E}\int_t^T \big|f(s, X_s^{t,x,\alpha}, \alpha_s)\big|\, ds < \infty. \tag{2.4}$$

  (If this is not automatic, it is made part of the definition of $\mathcal{A}_t$.)

Then:

(i) $w(t,x) \geq v(t,x)$.

(ii) If moreover there is a measurable $\hat{a} : [t,T) \times \mathbb{R}^d \to A$ such that

- **(V4) Equality along $\hat{a}$.** $\partial_s w + \mathcal{L}^{\hat{a}(s,y)} w + f(s,y,\hat{a}(s,y)) = 0$ for all (s,y) and $w(T,\cdot) = g$; i.e. w solves (2.1) and $\hat{a}$ attains the sup;
- **(V5) Admissible closed loop.** The closed-loop SDE (note 03 eq. 1.3) with $\hat{a}$ has a solution $\hat{X}$ started at (t,x), and $\hat{\alpha}_s = \hat{a}(s,\hat{X}_s)$ belongs to $\mathcal{A}_t$,

then $w(t,x) = v(t,x) = J(t,x;\hat{\alpha})$: w is the value function and $\hat{\alpha}$ is optimal.

How to read it. (i) says: any smooth supersolution of the HJB is an **upper bound** on what any admissible strategy can earn. (ii) says: if the bound is attained by an admissible feedback, the bound is the value. No a-priori knowledge of v (smoothness, DPP) is needed. The verification theorem uses only Itô's lemma.

2.3 Jump HJB

For the jump-controlled state of note 03 eq. (1.9), the generator acquires one term per counting process:

$$\mathcal{L}^a \varphi(t,x) = b(t,x,a)\,\partial_x \varphi + \tfrac{1}{2}\sigma^2(t,x,a)\,\partial_{xx}\varphi + \sum_{i=1}^{I} \lambda_i(t,x,a)\Big[\varphi\big(t, x+\eta_i(t,x,a)\big) - \varphi(t,x)\Big], \tag{2.5}$$

and the HJB is (2.1) with this $\mathcal{L}^a$:

$$\partial_t v + \sup_{a \in A}\Big\{\mathcal{L}^a v + f(\cdot,a)\Big\} = 0, \qquad v(T,\cdot) = g. \tag{2.6}$$

This is the controlled version of note 01 eq. (6.4) and note 02 eq. (5.1): "rate of the event × change in value if the event happens", where now the **controller chooses the rate**. Theorem 2.1 holds for (2.6) under one extra assumption (J), see §3.4.

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3. Derivation

3.1 Formal derivation of the HJB equation from the DPP

Roadmap. Two inequalities. (a) Using a *constant* control a on a short interval $[t, t+h]$ and the half (2.3) of note 03's DPP gives "$\partial_t v + \mathcal{L}^a v + f(a) \le 0$ for every a". (b) Using an almost-optimal control and the half (2.2) gives "$\partial_t v + \sup_a\{\cdots\} \ge 0$". Together: (2.1). Both steps are note 02 §3 with a sup.

Assumption (F) — heuristic. $v \in C^{1,2}([0,T)\times\mathbb{R})$ with $\partial_t v$, $\partial_x v$, $\partial_{xx} v$ bounded and continuous, (H1)–(H4) hold, and H is finite and continuous. **This is an assumption about the unknown v; it is generally not verifiable in advance and can fail** (Pitfall 1). That is why this subsection is called formal, and why note 05 (viscosity solutions) exists.

Step 1: Dynkin for a controlled process. For any $\alpha \in \mathcal{A}_t$, integrate Itô's lemma (Def. 1.1) on $[t, t+h]$ and take expectations. The dW integral has zero mean, because its integrand $\sigma(s, X_s, \alpha_s)\partial_x v$ has $\mathbb{E}\int |\cdot|^2 < \infty$ (bounded $\partial_x v$, growth (H2), moment bound note 03 eq. 1.4; note 01 §3). So

$$\mathbb{E}\big[v(t+h, X_{t+h})\big] = v(t,x) + \mathbb{E}\int_t^{t+h} \big(\partial_s v + \mathcal{L}^{\alpha_s} v\big)(s, X_s)\, ds. \tag{3.1}$$

Step 2 ("≤ 0"): constant controls. Fix $a \in A$ and use $\alpha_s \equiv a$, which is admissible. Note 03's (2.3) says $\mathbb{E}[\int_t^{t+h} f(s, X_s, a)\,ds + v(t+h, X_{t+h})] \le v(t,x)$. Substituting (3.1), the $v(t,x)$ cancels:

$$\mathbb{E}\int_t^{t+h} \Big[\big(\partial_s v + \mathcal{L}^a v\big)(s, X_s) + f(s, X_s, a)\Big]\,ds \le 0.$$

Divide by h and let $h \downarrow 0$. The integrand is continuous in (s, X_s), paths are continuous with $X_t = x$, and the integrand is bounded by $C(1 + \sup_s |X_s|^2)$, which is integrable (note 03 eq. 1.4). By dominated convergence the average tends to its value at (t,x):

$$\partial_t v(t,x) + \mathcal{L}^a v(t,x) + f(t,x,a) \le 0 \quad \text{for every } a \in A \quad \Longrightarrow \quad \partial_t v + H(t, x, \partial_x v, \partial_{xx} v) \le 0. \tag{3.2}$$

Step 3 ("≥ 0"): almost-optimal controls. By note 03's (2.2), for each h there is $\alpha^h \in \mathcal{A}_t$ with

$$v(t,x) - h^2 \le \mathbb{E}\Big[\int_t^{t+h} f(s, X_s^h, \alpha_s^h)\,ds + v(t+h, X_{t+h}^h)\Big],$$

where $X^h = X^{t,x,\alpha^h}$. Substituting (3.1) and then bounding the integrand pointwise by its supremum over control values (the definition (1.2) of H):

$$-h^2 \le \mathbb{E}\int_t^{t+h} \Big[\partial_s v + \mathcal{L}^{\alpha_s^h} v + f(\cdot, \alpha_s^h)\Big](s, X_s^h)\,ds \le \mathbb{E}\int_t^{t+h} G(s, X_s^h)\,ds, \tag{3.3}$$

with $G(s,y) := \partial_s v(s,y) + H(s, y, \partial_x v(s,y), \partial_{xx} v(s,y))$, a continuous function.

Divide by h: $-h \le \frac{1}{h}\mathbb{E}\int_t^{t+h} G(s, X_s^h)\,ds$. The controls α^h change with h, so we need $X_s^h \to x$ **uniformly in the control**. Since A is compact (H4), (H2) bounds b, σ by $C(1 + |X|)$ for every control, and the argument of note 03 Lemma 1.2 gives $\mathbb{E}\sup_{s \in [t, t+h]} |X_s^h - x|^2 \le C(1 + |x|^2)h$. Hence $G(s, X_s^h) \to G(t,x)$ in probability, uniformly over $s \in [t, t+h]$, and with dominated convergence

$$0 \le G(t,x) = \partial_t v(t,x) + H\big(t, x, \partial_x v, \partial_{xx} v\big). \tag{3.4}$$

Step 4: terminal condition. $v(T,x) = g(x)$ by definition (note 03 §1.4).

Combining (3.2) and (3.4) gives (2.1). $\square$ (formal)

Recap. Step 2 is "every action is at best optimal" and gives $\le$. Step 3 is "optimal actions exist approximately" and gives $\ge$. The sup in H appears in Step 3 when we bound $\mathcal{L}^{\alpha_s} v + f(\alpha_s)$ by its supremum over a. With a single control value, Steps 2 and 3 coincide and reproduce note 02 eq. (3.2).

> **Intuition:** $\partial_t v + \mathcal{L}^a v + f(a)$ is the expected rate of change of "value so far + value to go" (note 03 eq. 2.4) under action a. HJB says this rate is ≤ 0 for every action and $= 0$ for the best one: the infinitesimal form of "supermartingale for every control, martingale for the optimal one" (note 03 Cor. 2.2).

3.2 Feedback controls from the Hamiltonian

Given a solution w of (2.1), computing $\hat{a}$ in (2.2) is a static optimization at each (t,x).

Typical case (control in the drift, quadratic cost). $b = a$, σ independent of a, $f = -\frac{1}{2}a^2 + \ell(t,x)$, $A = \mathbb{R}$. The function $a \mapsto ap - \frac{1}{2}a^2$ is concave with maximizer $a = p$, so

$$\hat{a}(t,x) = \partial_x w(t,x), \qquad H = \tfrac{1}{2}\sigma^2 M + \tfrac{1}{2}p^2 + \ell.$$

The optimal drift **points up the gradient of the value function**, with speed proportional to its steepness.

What can go wrong (and must be checked in (V5)): the $\arg\max$ may not be unique; it may not be measurable or continuous in (t,x); and the closed-loop SDE may then fail to have a solution. In all examples below $\hat a$ is Lipschitz in x, so (V5) follows from note 01 Def. 4.2.

3.3 Proof of the verification theorem

Roadmap. Apply Itô to $w(s,X_s)$ for an arbitrary admissible control. (V2) says the drift of "$\int f\,ds + w(s,X_s)$" is ≤ 0, so its expectation can only decrease: this gives (i). Under $\hat\alpha$ the drift is exactly 0, which gives equality, i.e. (ii). The only technical work is that the dW integral might be just a local martingale (note 01, caution after eq. 3.2): we **localize** with stopping times, and remove the localization with dominated convergence, which is where (V1) and (V3) are used.

Step 0: localization. Fix $\alpha\in\mathcal{A}_t$, write $X=X^{t,x,\alpha}$, and for integers $n>1/(T-t)$ set

$$\tau_n=\inf\{s\ge t:|X_s-x|\ge n\}\wedge\big(T-\tfrac{1}{n}\big). \tag{3.5}$$

On $[t,\tau_n]$, (s,X_s) stays in the compact set $\mathcal{K}_n=[t,T-\frac1n]\times\{|y-x|\le n\}$, where $w,\partial_t w,\partial_x w,\partial_{xx}w$ are continuous (V1), hence bounded by some C_n. We stop before T because (V1) only gives $w\in C^{1,2}$ on $[0,T)$; at T the value function is often not smooth (e.g. g has a kink).

Step 1: Itô up to τ_n. By Itô's lemma (Def. 1.1) integrated from t to τ_n:

$$w(\tau_n,X_{\tau_n})=w(t,x)+\int_t^{\tau_n}\big(\partial_s w+\mathcal{L}^{\alpha_s}w\big)(s,X_s)\,ds+\int_t^{\tau_n}\sigma(s,X_s,\alpha_s)\,\partial_x w(s,X_s)\,dW_s. \tag{3.6}$$

Step 2: the stopped stochastic integral has zero mean. Write it as $\int_t^T \mathbf{1}_{\{s\le\tau_n\}}\sigma\,\partial_x w\,dW_s$; the indicator is adapted, so the integrand is adapted. Its square integral is finite:

$$\mathbb{E}\int_t^T \mathbf{1}_{\{s\le\tau_n\}}|\sigma\,\partial_x w|^2 ds\le C_n^2\,\mathbb{E}\int_t^T 3K^2\big(1+|X_s|^2+|\alpha_s|^2\big)ds<\infty,$$

using (H2), the moment bound note 03 eq. (1.4), and (1.2) there. So by note 01 §3 it is a true martingale with zero mean. (Without the localization we would need a growth bound on $\partial_x w$, which is not assumed.)

Step 3: use the supersolution inequality. Take expectations in (3.6). The ds integrand is bounded on $\mathcal{K}_n$ by $C_n(1+|b|+\sigma^2)$, which is integrable for the same reasons, so the expectation splits. By (V2) applied with $a=\alpha_s$ (pointwise in (s,ω)),

$$\big(\partial_s w+\mathcal{L}^{\alpha_s}w\big)(s,X_s)\le -f(s,X_s,\alpha_s).$$

Hence

$$w(t,x)\ \ge\ \mathbb{E}\Big[\int_t^{\tau_n} f(s,X_s,\alpha_s)\,ds+w(\tau_n,X_{\tau_n})\Big]. \tag{3.7}$$

Step 4: remove the localization. Let $n\to\infty$.

- $\tau_n\to T$ a.s.: the path $s\mapsto X_s$ is continuous on $[t,T]$, hence bounded, so for n large the first term in (3.5) is never reached and $\tau_n=T-\frac1n$.
- Running term: $\int_t^{\tau_n} f\,ds\to\int_t^T f\,ds$ a.s., and $|\int_t^{\tau_n} f\,ds|\le\int_t^T|f|\,ds$, integrable by (V3).

- Terminal term: $w(\tau_n, X_{\tau_n}) \to w(T, X_T)$ a.s., by continuity of w on $[0,T]\times\mathbb{R}^d$ (V1) and of the path; and $|w(\tau_n, X_{\tau_n})| \le C_w(1+\sup_s |X_s|^p)$, integrable by (V3).

By dominated convergence in (3.7), and then $w(T,\cdot)\ge g$ from (V2),

$$w(t,x) \ge \mathbb{E}\Big[\int_t^T f(s, X_s, \alpha_s)\,ds + w(T, X_T)\Big] \ge J(t,x;\alpha). \tag{3.8}$$

Since $\alpha \in \mathcal{A}_t$ was arbitrary, $w(t,x) \ge \sup_\alpha J = v(t,x)$. This proves **(i)**.

Step 5: equality for $\hat\alpha$. Repeat Steps 0–4 with $\alpha = \hat\alpha$ (admissible by (V5)). By (V4), the inequality in Step 3 is an equality, and $w(T,\cdot) = g$, so every "$\ge$" above is "$=$": $w(t,x) = J(t,x;\hat\alpha) \le v(t,x)$. With (i), $w(t,x) = v(t,x) = J(t,x;\hat\alpha)$. This proves **(ii)**. □

Recap: where each assumption was used.

Assumption	Used in
(V1) $C^{1,2}$ on $[0,T)$	Itô's lemma, Step 1; boundedness on $\mathcal{K}_n$, Steps 2–3
(V1) continuity on $[0,T]$	$w(\tau_n, X_{\tau_n}) \to w(T, X_T)$, Step 4
(V1) polynomial growth + (V3) moments	dominated convergence for the terminal term, Step 4
(V3) $\mathbb{E}\int \lvert f\rvert < \infty$	dominated convergence for the running term, Step 4
(H2) + note 03 (1.2), (1.4)	true martingale property of the stopped integral, Step 2
(V2)	the inequality, Step 3 and (3.8)
(V4), (V5)	equality and admissibility of the candidate optimum, Step 5

> **Remark (uniqueness).** If w_1, w_2 both satisfy (V1), (V4), (V5) (with (V3) for all controls), both equal v. So the HJB has **at most one** classical solution of polynomial growth with an admissible maximizer. Compare note 02 Pitfall 2 (Tychonoff): growth conditions select the right solution.

> **Remark (bounds without solving).** Part (i) needs only a *supersolution*. Any smooth w with $\partial_t w + H \le 0$, $w(T)\ge g$ is an upper bound on the value; any admissible control gives a lower bound J. This sandwich is often useful in practice.

3.4 Verification with jumps

Roadmap. Same proof, with Itô's lemma with jumps (note 01 eq. 6.3) in Step 1 and one more martingale in Step 2. We first state the Itô formula for counting processes with controlled intensity.

Itô with controlled intensities. Note 01's derivation of (6.3) was pathwise (between jumps use (5.1); at a jump, add the exact difference), so it holds for any counting process, whatever its intensity. For the state (note 03 eq. 1.9):

$$d\varphi(s,X_s) = \big(\partial_s\varphi + \mathcal{L}^{\alpha_s}\varphi\big)ds + \sigma\,\partial_x\varphi\,dW_s + \sum_i \Big[\varphi(s, X_{s-}+\eta_i) - \varphi(s,X_{s-})\Big]\big(dN^i_s - \lambda_i\,ds\big), \tag{3.9}$$

with $\mathcal{L}^a$ from (2.5) and η_i, λ_i evaluated at (s, X_{s-}, α_s). We wrote $dN^i = \lambda_i\,ds + (dN^i - \lambda_i\,ds)$; the $\lambda_i\,ds$ parts form the jump term of $\mathcal{L}^{\alpha_s}$.

Fact 3.1 (compensated integrals). If Z_s is predictable and $\mathbb{E}\int_t^T |Z_s|\lambda_i(s)\,ds < \infty$, then $\int_t^{\cdot} Z_s(dN_s^i - \lambda_i\,ds)$ is a martingale with zero mean (Brémaud, *Point Processes and Queues*, Ch. II, T8). Here $Z_s = \varphi(s, X_{s-} + \eta_i) - \varphi(s, X_{s-})$ is predictable because it uses X_{s-}.

Extra assumption (J). The intensities and jump sizes are bounded: $0 \le \lambda_i \le \bar\lambda$, $|\eta_i| \le \bar\eta$.

Proof of Theorem 2.1 for (2.6). Steps 0–5 go through with these changes:

- Step 1: use (3.9).
- Step 2: on $[t, \tau_n]$, X_{s-} lies in the ball of radius n around x and $X_{s-} + \eta_i$ in the ball of radius $n + \bar\eta$, where w is bounded. So $|Z_s| \le 2C_n$, and $\mathbb{E}\int |Z|\lambda \le 2C_n\bar\lambda T$. By Fact 3.1, the stopped compensated integrals have zero mean.
- Step 4: $\tau_n \to T$ still holds (a càdlàg path on $[t,T]$ is bounded), and $|X_{\tau_n}| \le |x| + n + \bar\eta$ is still controlled by $\sup_s |X_s|$, so dominated convergence applies as before. Left limits replace continuity: $w(\tau_n, X_{\tau_n}) \to w(T, X_{T-})$, and $X_{T-} = X_T$ a.s. because a jump exactly at the fixed time T has probability zero (bounded intensity). □

3.5 The market-making jump HJB

We now write (2.6) for the market-making state. It is used in Example 4.2 and in note 12.

State. Cash X_s, inventory $q_s \in \mathbb{Z}$, mid price S_s with $dS_s = \sigma\,dW_s$. The controls are the quote distances $\delta_s^b, \delta_s^a \in [0, \bar\delta]$: the bid is posted at $S_s - \delta_s^b$, the ask at $S_s + \delta_s^a$. Bid fills arrive at intensity $\Lambda(\delta_s^b)$ and ask fills at intensity $\Lambda(\delta_s^a)$, with $\Lambda(\delta) = Ae^{-k\delta}$. At a **bid fill** the market maker buys one unit at $S - \delta^b$: $(x, q) \to (x - s + \delta^b,\ q + 1)$. At an **ask fill** it sells one unit at $S + \delta^a$: $(x, q) \to (x + s + \delta^a,\ q - 1)$. So

$$dX_s = (S_s + \delta_s^a)\,dN_s^a - (S_s - \delta_s^b)\,dN_s^b, \qquad dq_s = dN_s^b - dN_s^a, \qquad dS_s = \sigma\,dW_s. \tag{3.10}$$

The intensities are bounded by A and the jump sizes in (x, q) are bounded on bounded s-sets, so (J) holds after the usual localization.

HJB. With a running reward $f = -\phi q^2$ (inventory penalty) and terminal reward $g(x, q, s)$, (2.5)–(2.6) give, for $v = v(t, x, q, s)$,

$$\partial_t v + \tfrac{1}{2}\sigma^2\partial_{ss}v - \phi q^2 + \sup_{\delta^b}\Lambda(\delta^b)\Big[v(t, x - s + \delta^b, q + 1, s) - v\Big] + \sup_{\delta^a}\Lambda(\delta^a)\Big[v(t, x + s + \delta^a, q - 1, s) - v\Big] = 0, \tag{3.11}$$

The sup over (δ^b, δ^a) **separates** into two one-dimensional problems, because each control affects only its own jump term. Avellaneda–Stoikov is (3.11) with $\phi = 0$ and $g = -e^{-\gamma(x+qs)}$ (CARA utility of note 01 Ex. 5.4); note 12 solves it.

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4. Worked examples

Example 4.1: LQ steering — fully solved

Roadmap. The problem of note 03 §4. We solve the HJB with a quadratic ansatz, reduce it to a Riccati ODE, solve that, and check (V1)–(V5). This upgrades note 03's lower bound (4.11) to an equality and proves the feedback $-P(t)X$ optimal.

The problem. $dX = \alpha\,ds + \sigma\,dW$, $A = \mathbb{R}$, $f = -\frac{1}{2}a^2$, $g = -\frac{1}{2}cx^2$ (note 03 eqs. 4.1–4.2).

HJB. By (1.1), $\mathcal{L}^a w = a\,\partial_x w + \frac{1}{2}\sigma^2 \partial_{xx} w$, so (2.1) reads

$$\partial_t w + \tfrac{1}{2}\sigma^2 \partial_{xx} w + \sup_{a \in \mathbb{R}} \left\{ a\,\partial_x w - \tfrac{1}{2}a^2 \right\} = 0, \qquad w(T,x) = -\tfrac{1}{2}cx^2. \tag{4.1}$$

By §3.2 the sup is attained at $\hat{a} = \partial_x w$ and equals $\frac{1}{2}(\partial_x w)^2$:

$$\partial_t w + \tfrac{1}{2}\sigma^2 \partial_{xx} w + \tfrac{1}{2}(\partial_x w)^2 = 0. \tag{4.2}$$

Ansatz. The terminal condition is quadratic and the dynamics are linear, so try

$$w(t,x) = -\tfrac{1}{2}P(t)\,x^2 - K(t), \qquad P(T) = c,\ K(T) = 0. \tag{4.3}$$

Then $\partial_x w = -Px$, $\partial_{xx} w = -P$, $\partial_t w = -\frac{1}{2}P'x^2 - K'$. Substitute in (4.2):

$$-\tfrac{1}{2}P'x^2 - K' - \tfrac{1}{2}\sigma^2 P + \tfrac{1}{2}P^2 x^2 = 0 \quad \text{for all } x.$$

Matching the x^2 and constant terms:

$$P' = P^2, \qquad K' = -\tfrac{1}{2}\sigma^2 P. \tag{4.4}$$

Solve. $(1/P)' = -P'/P^2 = -1$ and $1/P(T) = 1/c$, so $1/P(t) = 1/c + (T-t)$, i.e. $P(t) = \dfrac{c}{1 + c\tau}$. Then $K(t) = \frac{1}{2}\sigma^2 \int_t^T P(s)\,ds = \frac{1}{2}\sigma^2 \int_0^\tau \frac{c}{1+cu}\,du = \frac{1}{2}\sigma^2 \ln(1+c\tau)$. Hence

$$\boxed{w(t,x) = -\frac{c\,x^2}{2(1+c\tau)} - \frac{1}{2}\sigma^2 \ln(1+c\tau), \qquad \hat{a}(t,x) = -P(t)\,x = -\frac{c\,x}{1+c\tau}.} \tag{4.5}$$

This is the right side of note 03 eq. (4.11), and P is exactly the discrete P_n of note 03 eq. (4.9).

Verification (Theorem 2.1). Take $\mathcal{A}_t$ = progressively measurable real controls with $\mathbb{E}\int_t^T \alpha^2 ds < \infty$ (note 03 Def. 1.1).

- (V1): w is a quadratic polynomial in x with smooth coefficients on $[0,T]$; $p = 2$.
- (V2), (V4): by construction, $\partial_t w + \mathcal{L}^a w + f(a) = -\frac{1}{2}(a - \partial_x w)^2 \le 0$ for every a (complete the square), with equality iff $a = \hat{a}$; and $w(T) = g$.
- (V3): $X_s = x + \int_t^s \alpha + \sigma(W_s - W_t)$, so $\sup_s |X_s|^2 \le 3\big(x^2 + T\int \alpha^2 + \sigma^2 \sup_s |W_s - W_t|^2\big)$, integrable by Doob; and $\mathbb{E}\int |f| = \frac{1}{2}\mathbb{E}\int \alpha^2 < \infty$.
- (V5): the closed loop $d\hat{X} = -P(s)\hat{X}\,ds + \sigma\,dW$ is a linear SDE (Lipschitz), so it has a unique Gaussian solution; $\hat{\alpha} = -P\hat{X}$ has $\mathbb{E}\int \hat{\alpha}^2 < \infty$.

Therefore $v = w$ and $\hat{\alpha}_s = -P(s)X_s$ is optimal. Note 03's open-loop value (4.4) and discrete values (4.10) are strictly below v, as they must be by (i).

Note that this example violates (H4) ($A = \mathbb{R}$) and (H3) (quadratic rewards), which were needed for the DPP proof in note 03 and the formal derivation in §3.1. The verification theorem does not need them. This is a common pattern: derive HJB formally, then *prove* the answer by verification.

Numbers. $c = 4$, $\sigma = 1$, $T = 1$:

(t,x)	$P(t)$	$\hat{a}(t,x)$	$v(t,x)$
$(0, 1)$	0.8	-0.8	$-0.4 - \frac{1}{2}\ln 5 = -1.20472$
$(0.5, 1)$	$4/3$	-1.333	$-0.6667 - \frac{1}{2}\ln 3 = -1.21597$
$(0.9, 1)$	2.857	-2.857	$-1.4286 - \frac{1}{2}\ln 1.4 = -1.59681$

Comparison with suboptimal admissible controls (exact rewards from the moment ODE, see §5): doing nothing gives $-\frac{1}{2}c(x^2+\sigma^2 T) = -4$; the best **constant** gain $\alpha = -\kappa X$ is $\kappa \approx 1.561$ with reward -1.3443; both below $v = -1.2047$. The time-varying gain $P(t)$ matters.

Remark 4.1 (Cole–Hopf: the HJB is a heat equation in disguise). Put $w = \sigma^2 \ln \psi$. Then $\partial_x w = \sigma^2 \psi_x/\psi$, $\partial_{xx} w = \sigma^2(\psi_{xx}/\psi - \psi_x^2/\psi^2)$, $\partial_t w = \sigma^2 \psi_t/\psi$. Substituting into (4.2), the ψ_x^2 terms cancel:

$$\frac{\sigma^2}{\psi}\Big(\partial_t \psi + \tfrac{1}{2}\sigma^2 \partial_{xx}\psi\Big) = 0, \qquad \psi(T,x) = e^{-cx^2/(2\sigma^2)}. \tag{4.6}$$

So ψ solves the backward heat equation, and Feynman–Kac (note 02 eq. 1.3) gives $\psi(t,x) = \mathbb{E}\big[e^{-c(x+\sigma W_\tau)^2/(2\sigma^2)}\big]$. With the Gaussian identity $\mathbb{E}[e^{-\beta(m+sZ)^2}] = (1+2\beta s^2)^{-1/2} e^{-\beta m^2/(1+2\beta s^2)}$ for $Z \sim \mathcal{N}(0,1)$, $\beta = c/(2\sigma^2)$, $m = x$, $s^2 = \sigma^2\tau$ (so $2\beta s^2 = c\tau$):

$$\psi = (1+c\tau)^{-1/2} \exp\Big(-\frac{cx^2}{2\sigma^2(1+c\tau)}\Big) \quad \Longrightarrow \quad \sigma^2 \ln\psi = w \text{ of (4.5). } \checkmark$$

This independent route (linear PDE + expectation) is one of the numerical checks.

> **Intuition:** the optimal controller pays $\frac{1}{2}\sigma^2 \ln(1+c\tau)$ for noise, against $\frac{1}{2}c\sigma^2\tau$ for the open-loop one. The logarithm is the signature of continuously "re-hedging" noise as it arrives; the linear term is what happens when you let it accumulate.

Example 4.2: risk-neutral market maker — jump HJB

Roadmap. Solve (3.11) with $\phi = 0$ and terminal reward $g = x + qs$ (marked-to-mid wealth, no risk aversion). An ansatz removes x and s, the jump terms become explicit, and each sup is a 1-d calculus problem. We recover note 02 Ex. 5.3's $\delta^* = 1/k$, now as a proved optimal control. Then we add $\phi > 0$ to see inventory skew appear.

Ansatz. Try

$$w(t,x,q,s) = x + q\,s + \theta(t,q), \qquad \theta(T,q) = 0. \tag{4.7}$$

Then $\partial_{ss} w = 0$, and the jump differences are

$$w(t, x - s + \delta^b, q+1, s) - w = -s + \delta^b + s + \theta(t,q+1) - \theta(t,q) = \delta^b + \theta(t,q+1) - \theta(t,q),$$

$$w(t, x + s + \delta^a, q-1, s) - w = s + \delta^a - s + \theta(t,q-1) - \theta(t,q) = \delta^a + \theta(t,q-1) - \theta(t,q).$$

The mid price cancels: a fill at $S \mp \delta$ is worth exactly δ relative to marking at mid. Substituting in (3.11), all dependence on x, s drops out, and θ solves the **ODE system indexed by** $q \in \mathbb{Z}$

$$\partial_t \theta(t,q) - \phi q^2 + \sup_{\delta^b} \Lambda(\delta^b)\big[\delta^b + \theta(t,q+1) - \theta(t,q)\big] + \sup_{\delta^a} \Lambda(\delta^a)\big[\delta^a + \theta(t,q-1) - \theta(t,q)\big] = 0. \tag{4.8}$$

(Compare note 02 eq. (5.2): integer state $\Longrightarrow$ system of ODEs.)

The pointwise sup. For $p \in \mathbb{R}$, maximize $F(\delta) = Ae^{-k\delta}(\delta + p)$ over $\delta \geq 0$. $F'(\delta) = Ae^{-k\delta}\big(1 - k(\delta+p)\big)$, positive for $\delta < 1/k - p$ and negative after, so

$$\delta^*(p) = \Big(\frac{1}{k} - p\Big) \vee 0, \qquad \sup_{\delta \geq 0} F = \frac{A}{e\,k}\, e^{kp} \quad \text{if } p \leq \frac{1}{k}. \tag{4.9}$$

(At the interior optimum, $\delta^* + p = 1/k$ and $e^{-k\delta^*} = e^{-1}e^{kp}$.) The optimal bid distance is $\delta^{b*} = \frac{1}{k} - \big[\theta(q+1) - \theta(q)\big]$: the quote moves by exactly the marginal value of one more unit of inventory.

Case $\phi = 0$. Try $\theta = \theta(t)$, independent of q. Then both p's are 0, both quotes are $\delta^* = 1/k$, both sups equal $A/(ek)$, and (4.8) becomes $\theta' + 2A/(ek) = 0$, $\theta(T) = 0$:

$$\boxed{v(t,x,q,s) = x + qs + \frac{2A}{e\,k}\,(T-t), \qquad \delta^{b*} = \delta^{a*} = \frac{1}{k}.} \tag{4.10}$$

Verification. Control set $[0, \bar\delta]$ with $\bar\delta > 1/k$ (so the optimum is interior); intensities $\le A$ and the jump in (x, q) is bounded on bounded s-sets, so (J) holds after localization. (V1): w is smooth and of linear growth ($p = 1$). (V3): total fills on $[t, T]$ are dominated by a Poisson$(2A\tau)$ variable, which has all moments; S is Gaussian; so q, X, S have all moments, and $f = 0$. (V2), (V4): by (4.9) with $p = 0$, sup is attained at $1/k$. (V5): constant quotes are admissible. Hence (4.10) is the value function and constant quotes at $1/k$ are optimal.

Numbers. Avellaneda–Stoikov parameters $A = 140$, $k = 1.5$, $T = 1$: $\delta^* = 0.6667$ and $\frac{2A}{ek}T = \frac{280}{1.5\,e} = 68.671$. That is $2 \times 140e^{-1} = 103.0$ expected fills, each earning 0.6667.

Case $\phi > 0$ (inventory aversion, numerical). The constant-in-q ansatz fails because of $-\phi q^2$, but (4.8)–(4.9) is still a closed ODE system, solved numerically in `code/hjb_checks.py` (truncated at $|q| \le 60$). With $\phi = 0.05$:

q	-2	-1	0	$+1$	$+2$
$\delta^{b*}(0,q)$	0.6283	0.6539	0.6795	0.7051	0.7307
$\delta^{a*}(0,q)$	0.7307	0.7051	0.6795	0.6539	0.6283

$\theta(0,0) = 67.595$. A long market maker ($q > 0$) **widens its bid and tightens its ask**, so it is more likely to sell than buy: the inventory skew of note 01 Ex. 5.4, now produced by the HJB. Keeping symmetric quotes at $1/k$ earns only ≈ 66.08 under the same penalty (Monte Carlo), consistent with (i) of Theorem 2.1. The skew per unit of inventory here is ≈ 0.0256; notes 12–13 derive its closed-form approximations.

> **Intuition:** in (4.9) the quote is "monopoly half-spread $1/k$ minus the marginal value of the inventory change the fill causes". If buying one more unit lowers your continuation value ($\theta(q+1) < \theta(q)$), you demand a bigger discount to buy.

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5. Numerical check

Script: `code/hjb_checks.py`; tests: `code/tests/test_04_hjb.py` (run `python -m pytest code/tests -q`).

Check	Methods compared	Expected	Tolerance
Riccati	`solve_ivp` on (4.4) vs closed form	$P(0) = 0.8$, $K(0) = \frac{1}{2}\ln 5$	10^{-9}
HJB residual	finite differences of (4.5) plugged into (4.2) at 9 points	0	10^{-5}
HJB PDE solve	explicit monotone FD scheme for (4.1), brute-	-1.20472	10^{-3} (observed $2.6 \cdot 10^{-4}$, second order)

Check	Methods compared	Expected	Tolerance
	force sup over 401 control values, no ansatz		
Cole–Hopf	Monte Carlo of $\sigma^2 \ln \mathbb{E}[e^{-cX_T^2/2\sigma^2}]$ (Remark 4.1)	-1.20472	$3 \cdot 10^{-3}$
Verification (i)	exact rewards of constant-gain feedbacks via moment ODE, and their optimum	all $< v$; best -1.3443	strict
Sup formula	(4.9) vs brute-force grid over δ	—	10^{-7} rel.
Risk-neutral MM	ODE system (4.8) vs (4.10) vs Monte Carlo of quotes at $1/k$	68.671, $\delta^* = 0.6667$	10^{-6}; 4 s.e.
Inventory penalty	ODE with closed-form sup vs explicit solver with grid sup vs Monte Carlo of the resulting feedback; skew monotone; symmetric quotes worse	$\theta(0,0) = 67.595$	10^{-3}; 4 s.e. + 0.02

The FD scheme uses central differences for the drift; it is monotone because $|a| \le \sigma^2/\Delta x$ on the control grid and $\Delta t \le \Delta x^2/\sigma^2$. Monotonicity is the property that guarantees convergence to the right (viscosity) solution — the Barles–Souganidis theorem of note 05.

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6. Pitfalls / intuition

1. **The HJB may have no classical solution.** Take $dX = \alpha\, ds$ (no noise), $A = [-1, 1]$, $f = 0$, $g(x) = -|x|$. Moving toward 0 at full speed is optimal, so $v(t,x) = -(|x| - \tau)^+$. It is not differentiable on $|x| = \tau$. The HJB here is $\partial_t v + |\partial_x v| = 0$, and v satisfies it only almost everywhere.
2. **"Almost-everywhere solutions" are not unique.** For the same dynamics with $g = 0$, the true value is $v \equiv 0$, but $\tilde{w}(t,x) = (\tau - |x|)^+$ also satisfies $\partial_t \tilde{w} + |\partial_x \tilde{w}| = 0$ a.e. (on $|x| < \tau$: $-1 + 1 = 0$) and $\tilde{w}(T) = 0$. So the PDE in the a.e. sense does not determine v. Viscosity solutions (note 05) fix this.
3. **The Hamiltonian can be infinite.** With unbounded controls and no cost of control, $H = +\infty$ and the problem is degenerate (arbitrarily large rewards, or a singular/impulse control problem; note 14).
4. **Local martingales.** Without localization and integrability, the "$\mathbb{E} \int \cdots dW = 0$" step can fail. Doubling-type strategies are admissible-looking controls that break it; (V3) and (1.2) of note 03 exclude them.
5. **Check the closed loop.** An $\arg\max$ that is discontinuous in x (bang-bang controls, as in Pitfall 1) can make the closed-loop SDE ill-posed. (V5) is a real condition.
6. **Formal vs proved.** §3.1 is heuristic (it assumes v smooth). Theorem 2.1 is a full proof, modulo the quoted Fact 3.1 for jumps. In practice: guess from §3.1, prove by Theorem 2.1, or fall back on viscosity theory + numerics when no smooth solution exists.

7. **Textbook vs practice.** Real market makers do not solve (3.11) online. They use the structure it reveals — quotes = base half-spread − marginal inventory value — with parameters (A, k, ϕ) estimated and recalibrated from data (note 15).

中文小结

- **HJB** 方程：由 DPP 加 Itô/Dynkin 形式推导：用常数控制得到"对每个 a，$\partial_t v + \mathcal{L}^a v + f \le 0$"，用近似最优控制得到"$\ge 0$"，合起来得 $\partial_t v + \sup_a\{\mathcal{L}^a v + f\} = 0$，$v(T) = g$。这是第 02 篇线性 PDE 加一个逐点 sup；推导假设 v 光滑，属于启发式。
- **Hamiltonian**：$H = \sup_a\{bp + \frac{1}{2}\sigma^2 M + f\}$，关于 (p, M) 凸且对 M 单调（退化椭圆性）。最优反馈 $\hat{a}(t,x)$ 是逐点最大化的解，把对过程的无穷维优化变成每个点的有限维优化。
- 验证定理（完整证明）：光滑、多项式增长的上解 w 是任何容许策略收益的上界；若 $\hat{a}$ 达到 sup 且闭环 SDE 容许，则 $w = v$，$\hat{a}$ 最优。证明用停时局部化保证随机积分是真鞅，再用控制收敛去掉局部化；每个假设的用途都列在表中。
- 带跳 **HJB**：生成元加上 $\sum \lambda_i(a)[v(x+\eta_i) - v(x)]$，控制者选择跳跃强度；做市问题中买卖两侧的 sup 可分离。Avellaneda–Stoikov 就是 $\phi = 0$、CARA 终端效用的情形（第 12 篇）。
- 例 **1**（**LQ**）：二次猜解化为 Riccati $P' = P^2$，$v = -\frac{cx^2}{2(1+c\tau)} - \frac{1}{2}\sigma^2 \ln(1+c\tau)$，反馈 $-P(t)x$；$c = 4, \sigma = 1, x = 1$ 时 $v = -1.2047$。Cole–Hopf 变换把 HJB 化为热方程，可用 Feynman–Kac 独立验证。
- 例 **2**（风险中性做市）：猜解 $x + qs + \theta$，最优报价 $1/k$，$v = x + qs + \frac{2A}{ek}\tau$；$A = 140, k = 1.5$ 时为 68.67。加入库存惩罚 ϕq^2 后，多头时买价变宽、卖价变窄（库存偏移），数值解与蒙特卡洛一致。
- 陷阱：HJB 可能没有经典解，几乎处处解不唯一（需要粘性解，第 05 篇）；Hamiltonian 可能为无穷；必须检查局部鞅与闭环 SDE 的适定性。